

1. Administrative & Study Identification

Official Study Title A Phase III Randomized, Open-Label Study Evaluating Efficacy and Safety of Giredestrant Compared With Fulvestrant, Both Combined With a CDK4/6 Inhibitor, in Patients With Estrogen Receptor-Positive, HER2-Negative Advanced Breast Cancer With Resistance to Prior Adjuvant Endocrine Therapy **Brief Study Title** A Study to Evaluate Efficacy and Safety of Giredestrant Compared With Fulvestrant (Plus a CDK4/6 Inhibitor), in Participants With ER-Positive, HER2-Negative Advanced Breast Cancer Resistant to Adjuvant Endocrine Therapy (pionERA Breast Cancer) **Short Title/Acronym** pionERA Breast Cancer (BC) **Protocol Number** C044657 (2022-502980-39-00) **NCT Number** NCT06065748 **Sponsor/Company Name** Roche (F. Hoffmann-La Roche Ltd.) **Investigational Product** Giredestrant (oral selective estrogen receptor degrader [SERD]) **Phase of Development** Phase III **Trial Status** Recruiting **Medical Monitor** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (Progression-Free Survival [PFS]) of giredestrant in combination with a CDK4/6 inhibitor compared to fulvestrant in combination with a CDK4/6 inhibitor in patients with ER+/HER2- locally advanced or metastatic breast cancer (LA/mBC) resistant to prior adjuvant endocrine therapy. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) in the ESR1 no-mutation-detected (ESR1nmd) Subgroup, Overall Survival (OS), and Confirmed Objective Response Rate (cORR), along with exploratory assessments for Duration of Response (DoR), Clinical Benefit Rate (CBR), time to chemotherapy, safety, and patient-reported outcomes.

2.2 Background & Rationale Patients with ER+, HER2- LA/mBC who relapse on or after adjuvant endocrine therapy (ET) have a high unmet medical need. While fulvestrant plus a CDK4/6 inhibitor is a standard of care in the 1L setting for endocrine-resistant disease, fulvestrant requires monthly intramuscular injections that negatively impact quality of life (QoL) and has limited efficacy in patients with ESR1-mutated (ESR1m) tumors. Giredestrant is a highly potent, oral SERD that achieves robust ER occupancy regardless of ESR1m status. This study aims to determine whether replacing fulvestrant with giredestrant improves survival, QoL, tolerability, and adherence in this population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Fulvestrant) **Blinding:** Open-label

Randomization: 1:1, stratified by: prior adjuvant CDK4/6i (yes vs. no), choice of CDK4/6i, ESR1m status by central testing (ESR1m vs. no ESR1 mutation detected), and disease site (visceral vs. non-visceral).

3.2 Planned Enrollment & Duration Targeted Enrollment: 1,050 patients **Number of Sites:** Global, multicenter study (~38+ countries) **Estimated Study Start:** 10-Dec-2023 **Estimated Primary Completion:** 11-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Locally advanced or metastatic adenocarcinoma of the breast, not amenable to treatment with curative intent. 2. Documented estrogen receptor-positive (ER+), HER2-negative (HER2-) tumor assessed locally on the most recent tumor biopsy (or an archived tumor sample if a recent tumor sample is not available for testing). 3. Confirmed ESR1 mutation status (ESR1m versus ESR1nmd) in baseline circulating tumor DNA (ctDNA) through central laboratory testing. 4. Resistance to prior adjuvant endocrine therapy (ET), which is defined as having relapsed with prior standard adjuvant ET, on-treatment after $\geq$ 12 months or off-treatment within 12 months of completion. Prior use of adjuvant CDK4/6i is allowed (if relapse occurred $\geq$ 12 months since completion). 5. No prior systemic anti-cancer therapy for advanced disease. 6. Measurable disease as defined per RECIST v.1.1 or non-measurable (including bone-only) disease. 7. Eastern Cooperative Oncology Group Performance Status (ECOG PS) 0-1. 8. For pre/perimenopausal women and for men: willing to undergo and maintain treatment with approved LHRH agonist therapy (as per local guidelines) for the duration of study treatment.

4.2 Exclusion Criteria 1. Prior systemic therapy (e.g., prior chemotherapy, immunotherapy, or biologic therapy) for locally advanced unresectable or metastatic breast cancer. 2. Prior treatment with another SERD (e.g., fulvestrant, oral SERDs) or novel ER-targeting agents. 3. Advanced, symptomatic, visceral spread that is at risk of life-threatening complications in the short term. 4. Active cardiac disease or history of cardiac dysfunction. 5. Clinically significant history of liver disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** Giredestrant 30 mg once daily (Days 1-28) + Investigator's choice of CDK4/6 inhibitor (Abemaciclib 150 mg BID Days 1-28, Palbociclib 125 mg QD Days 1-21, or Ribociclib 600 mg QD Days

1-21). * **Active Comparator Arm:** Fulvestrant 500 mg intramuscularly (Days 1 and 15 of Cycle 1, then Day 1 of each subsequent cycle) + Investigator's choice of CDK4/6 inhibitor. **Route of Administration:** Oral (Giredestrant, CDK4/6i) and Intramuscular (Fulvestrant). **Duration of Treatment:** 28-day treatment cycles, continuing until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Luteinizing hormone-releasing hormone (LHRH) agonist therapy is required for pre/perimenopausal female participants and male participants on Day 1 of each cycle. **Prohibited:** Other concurrent anti-cancer therapies, other SERDs, or novel ER-targeting agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Central ESR1m testing, Baseline Tumor Assessment
Baseline	Day 0	Randomization, First Dose
Interim	Day 1 of Every Cycle	Safety Labs, Efficacy Assessment (RECIST v1.1), Oral Medication Accountability
End of Study	Variable	Disease Progression, Final Vital Signs, Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Progression-Free Survival (PFS) in the ESR1 mutation (ESR1m) Subgroup:** PFS is defined as the time from randomization to the first occurrence of PD, as determined by the investigator according to Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST v1.1), or death from any cause during the study. 2. **PFS in the Full Analysis Set (FAS) Population.**

7.2 Secondary & Exploratory Outcomes 1. **PFS in the ESR1 no-mutation-detected (ESR1nmd) Subgroup** 2. **Overall Survival (OS)** 3. **Confirmed Objective Response Rate (cORR)** 4. Exploratory: Duration of Response (DoR), Clinical Benefit Rate (CBR) and Time to chemotherapy 5. Exploratory: Patient-Reported Outcomes (PROs) and Safety

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical evaluations, lab assessments, and patient-reported outcomes at each 28-day cycle visit. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (iDMC) will be in place to monitor patient safety throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set (FAS) population and the ESR1m Subgroup for primary efficacy outcomes. **Sample Size**

Justification: Target enrollment of 1,050 patients to adequately power the detection of PFS and OS improvements. The co-primary endpoints will be compared between treatment arms using a stratified log-rank test at an overall 0.05 significance level (two-sided). **Handling of Missing Data:** Standard right-censoring techniques for time-to-event variables (PFS, OS, DoR).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure protocol compliance and data integrity. **Audit Trail:** Robust eCRF tracking, centralized laboratory data integration (for ESR1m testing), and standard data clarification form processes.

11. Study Identifiers & Governance

Primary ID: C044657 **Registry Number:** NCT06065748 **Sponsor/Lead Organization:** Roche (F. Hoffmann-La Roche Ltd.) **Study Title:** pionERA Breast Cancer (BC) **Official Regulatory Title:** A Phase III Randomized, Open-Label Study Evaluating Efficacy and Safety of Giredestrant Compared With Fulvestrant, Both Combined With a CDK4/6 Inhibitor, in Patients With Estrogen Receptor-Positive, HER2-Negative Advanced Breast Cancer With Resistance to Prior Adjuvant Endocrine Therapy

12. Clinical Framework (Breast Cancer Specific)

Primary Condition / Disease Area: Estrogen Receptor-Positive, HER2-Negative Advanced Breast Cancer **Diagnosis Threshold:** ER+/HER2- status with confirmed resistance to prior adjuvant endocrine therapy **Medical Methodology:** Targeted Endocrine Therapy + CDK4/6 Inhibition **Optimization Target:** ER-mediated signaling inhibition and sustained tumor suppression

13. Study Procedures & Methodology

Management Pathway: First-line (1L) therapy for relapsed/metastatic setting following adjuvant resistance. **Education Protocol (HCP):** Site initiation visits covering RECIST v1.1 tumor assessment and central laboratory ESR1 testing protocols. **Education Protocol (Patient):** Patient diary for oral giredestrant

and CDK4/6i compliance, QoL questionnaires. **Quality Audit Mechanism:** Centralized biomarker testing and ongoing iDMC safety reviews.

14. Observation & Follow-Up Schedule

Total Duration: Up to ~62 months (Dec 2023 - Feb 2029) **Onsite Visit Cadence:** Day 1 of every 28-day cycle **Remote Monitoring Frequency:** As per institutional/sponsor standards **Checklist Review Interval:** Radiographic tumor assessments typically conducted every 8-12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				